Praveen Rangavajhula

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Research Interests

Ph.D. student in Computer Science at the University of Georgia interested in how large language and multimodal models learn, represent information, and reason. My research interests span learning dynamics, internal representations, and the relationship between representations and model behavior, explored through controlled experiments, representation analysis, and mechanistic interpretability.

Education

- **University of Georgia (UGA)** — *Athens, GA*
  Ph.D. in Computer Science — *Jan 2026 - Present*
- **University of Georgia (UGA)** — *Athens, GA*
  M.S. in Computer Science (GPA: 3.91) — *Aug 2023 - Dec 2025*
- **BITS Pilani** — *Pilani, India*
  B.E. in Mechanical Engineering — *Aug 2019 - July 2023*

Research Experience

- **Graduate Research Assistant** — *University of Georgia*
  Longitudinal Machine Learning — *Feb 2026 - Present*

Cognitive Development Prediction (FFCWS and MIDUS)

- **End-to-End ML Pipeline**: Owned the complete machine learning pipeline for predicting cognitive outcomes from longitudinal Future of Families and Child Wellbeing Study (FFCWS) data, including data ingestion, cross-wave alignment, preprocessing, train-only imputation, model implementation, evaluation, analysis, and visualization for 1,480 children across ages 3, 5, and 9.
- **Model Development and Benchmarking**: Implemented and benchmarked nine model families: standard and feature-engineered linear regression, support vector regression, Random Forest, XGBoost, histogram gradient boosting, an attention network, TabICLv2, and a custom Neural CDE. Evaluated the models using five repeated child-level 80/20 train-test splits and 344 to 796 time-valid predictors.
- **Empirical Findings**: Found that expanding from theory-driven features to hundreds of predictors increased TabICLv2's average held-out R^2 from 0.307 to 0.324, indicating limited incremental predictive value from the expanded feature space.
- **Cross-Dataset Extension**: Extending the end-to-end pipeline to the Midlife in the United States (MIDUS) longitudinal study to examine whether the findings generalize to an independent cohort.

Language Models for Substance Use Prediction — *Aug 2026 - Present*

- **Survey-to-Text Dataset for Substance-Use Prediction**: Built and verified two text representations of FFCWS survey histories for 2,847 participants and 403 predictors collected through Year 15, targeting the Year-22 count of five substance-use categories. Decoded survey responses using official documentation, preserved missingness and survey skips, and excluded identifiers and outcome information from the inputs.
- **LLM Adaptation**: Running experiments with DeepSeek-V3 and LoRA adaptation to model longitudinal survey histories and predict young-adult substance-use outcomes.
- **Comparative Evaluation**: Comparing conventional tabular baselines, frozen language-model representations with Ridge regression, and LoRA approaches using regression heads, count-token prediction, and Number Token Loss to investigate the predictive value of survey meaning and temporal context.

- **Master's Researcher** — *University of Georgia*
  Advised by Prof. John A. Miller — *Aug 2024 - Dec 2025*

- **Custom Autograd Engine**: Designed and implemented a custom autograd engine in Scala, enabling automatic computation graph construction and backpropagation.
- **Gradient Caching for Efficiency**: Integrated gradient caching mechanisms to store intermediate values, reducing redundant recomputation and improving training efficiency.
- **Dynamic Computation Graph**: Implemented a dynamic graph structure that supports complex tensor operations, allowing for efficient forward and backward passes in deep learning models.
- **Scalability and Modularity**: Developed reusable tensor and optimization modules, making the engine extensible for future integration with larger ML frameworks, including potential applications in large-scale model training.
- **RNN and GRU Forecasting**: Manually derived backpropagation through time (BPTT) equations for RNNs and GRUs, analyzed gradient flow behavior, and applied gradient clipping to improve training stability.
- **Implementation in Scala**: Implemented RNNs and GRUs from scratch using the derived equations, achieving a SMAPE reduction from 60% to 16% on forecasting tasks.
- **Recurrent Architecture**: Designed modular `RecurrentBase` and `RNNCell` components for large-scale forecasting tasks.

- **Volunteer Researcher** *University of Georgia*
  Advised by Profs. Fei Dou and Lakshmish Ramaswamy *Jan 2025 - May 2025*
  - **Federated Learning for Carbon Footprint Tracking**: Implemented **Federated Learning** using **Flower** and **PyTorch** to track **carbon emissions of deep learning models (MobileNetV3, ResNet20)** across distributed nodes.
  - **Carbon-Aware Model Evaluation**: Analyzed trade-offs between **model accuracy, energy efficiency, and environmental impact** in federated deep learning settings.
  - **Edge AI Experimentation**: Conducted experiments on **Jetson devices** to evaluate federated learning models (MobileNetV3, ResNet20) under low-power, decentralized settings.

Skills

- **Programming Languages**: Python, Scala, Java, C, SQL
- **Machine Learning Libraries**: PyTorch, TensorFlow, Scikit-Learn, OpenCV
- **Data Science Tools**: NumPy, Pandas, Matplotlib, Seaborn, Jupyter, scikit-image
- **ML Techniques & Concepts**: Supervised Learning, Deep Learning, Representation Learning, Video Representation Learning, Latent-Space Prediction, Optimization, Backpropagation, Federated Learning, Differential Privacy, Autograd Systems

Projects

- **Visual Grounding in MedGemma** *Independent / Course Research*
  Advanced Representation Learning (CSCI 8945) *Aug 2026 - Present*
  - **Research Focus**: Investigating how MedGemma 1.5 uses image and text representations to answer chest X-ray questions, with an initial focus on pleural effusion.
  - **SAE Transfer and Layer Analysis**: Implemented Gemma Scope 2 sparse autoencoder reconstruction and activation-capture workflows for MedGemma's language decoder, using Gemma 3 as a reference control and screening all 34 decoder layers on a fixed 20-case CheXpert pilot.
  - **Feature Inspection**: Built a saved-activation pipeline and local viewer to compare sparse features across X-rays, expert labels, model answers, and image and text token positions as a foundation for further representation analysis.
- **Adversarial Audio Attacks on Voice Assistants** *Course Project, Prof. Jian Liu*
  Security and Privacy in Intelligent Audio Systems (CSCI 8000) *Aug 2026 - Present*
  - **Controlled Comparative Study**: Investigating adversarial attack effectiveness and transferability between direct audio input to Qwen2.5-Omni-3B and a cascaded Whisper-to-Qwen assistant, using the same frozen Qwen checkpoint. Testing attacks adapted to each route against both pipelines.
  - **Direct Audio Pipeline and Attack Adaptation**: Developing the direct audio inference pipeline and adapting an existing adversarial audio attack. Established the shared Python repository, reproducible uv dependency environment, and component structure for inference, attacks, and evaluation.
  - **Cross-Pipeline Evaluation**: Comparing final assistant responses across the four attack/pipeline conditions and inspecting intermediate transcripts to analyze attack effectiveness and transferability.
- **Predictive Video Representation Learning with V-JEPA 2** *Research Project*
  Video Future Selection and Latent Prediction *Jan 2026 - May 2026*
  - **V-JEPA Integration and Latent Prediction**: Built a JEPA-style video future-selection system using a frozen V-JEPA 2 encoder to extract normalized clip embeddings and a compact PyTorch latent predictor trained with a composite MSE and cosine loss to forecast next-step representations.
  - **End-to-End Data and Training Pipeline**: Developed MovingMNIST and real-video ingestion pipelines, standardized clips for the video encoder, generated K candidate futures per example, cached extracted embeddings, and implemented configurable AdamW training with validation-based best-state selection.
  - **Scoring, Baselines, and Evaluation**: Implemented frozen V-JEPA, learned latent-predictor, semantic, heuristic, and frozen-feature reranking methods; produced candidate rankings, confidence margins, and reproducible CSV/JSON benchmarks that revealed dataset-dependent trade-offs among scorers.
  - **Grounded Interpretability and Demonstration**: Created a deterministic explanation layer that converts structured scoring evidence into human-readable commentary, LLM-ready presenter prompts, and visualization-ready summaries for Jupyter demonstrations and research presentations.

- **Representation Learning Under Alternative Learning Rules** *University of Georgia*
  Independent Course Project in Representation Learning *Jan 2026 - May 2026*
  - **Learning Rule Implementations**: Independently implemented backpropagation, predictive coding, and Mono-Forward under matched multilayer perceptron architectures, conducting controlled experiments and validation-based model selection on MNIST and CIFAR-10.
  - **Representation Analysis**: Analyzed learned representations using layer-wise linear probes, t-SNE, and linear centered kernel alignment (CKA); Mono-Forward achieved 98.38% test accuracy on MNIST, 54.84% on CIFAR-10, and the strongest probe performance across both datasets.
  - **Representation Similarity**: Quantified learning-rule effects on representation geometry: backpropagation and Mono-Forward achieved a mean same-depth CKA of 0.749 on CIFAR-10, compared with 0.411 and 0.425 for pairings involving predictive coding.
  - **Memory Profiling**: Profiled peak CPU memory across matched MLPs with 2 to 10 hidden layers, observing lower measured peak memory for Mono-Forward as network depth increased.
- **Differentially Private Optimizer Research Project** *University of Georgia*
  Course Project in Privacy-Preserving Machine Learning (under Prof. Jaewoo Lee) *Aug 2024 - Dec 2024*
  - **DP-Lion: Private Optimizer Design**: Developed **DP-Lion**, a novel differentially private optimizer for deep learning, improving training stability and performance over DP-SGD and DP-Adam.
  - **Benchmarking on CIFAR-10**: Achieved **69% accuracy** on ResNet20 with privacy guarantees, outperforming most existing DP optimizers in empirical evaluations.
  - **Evaluation Against DeepMind Benchmarks**: Integrated techniques from DeepMind's private optimization research and compared DP-Lion against their reported baselines.
  - **Research Focus**: Used Opacus to privatize DP-Lion and baseline optimizers (DP-SGD, DP-Adam, DP-RMSprop), conducting empirical comparisons as part of a graduate privacy-preserving ML course project.
  - **Code**: GitHub Repository
- **Spotify Song Genre Classification** *University of Georgia*
  Course Project in Data Mining (under Prof. Fei Dou) *Aug 2024 - Dec 2024*
  - **Data Processing & Feature Engineering**: Processed **114,000 song samples**, extracting insightful **audio features** (e.g., **danceability-valence correlation**).
  - **Model Development**: Trained and compared **Decision Trees, Random Forests, SVM, kNN, and Neural Networks**, achieving **85.6% accuracy**.
  - **Experimentation & Metrics**: Designed and evaluated experiments using **precision-recall, ROC-AUC**, and **confusion matrices** to optimize classification.
  - **Code**: GitHub Repository
- **Diabetes Progression Prediction** *Kaggle Project*
  Data-Driven Modeling & Predictive Analytics *Sept 2024*
  - **Exploratory Data Analysis (EDA)**: Performed **data cleaning, feature engineering, and transformation (Box-Cox)** to improve regression model performance.
  - **Predictive Modeling**: Built and evaluated a **Linear Regression model from scratch**, benchmarking against **Scikit-Learn** implementations.
  - **Model Evaluation & Diagnostics**: Used **residual analysis, Q-Q plots, VIF checks**, and **homoscedasticity tests** to diagnose and improve model performance.
  - **Regularization & Feature Selection**: Analyzed the impact of **Lasso, Ridge, and Elastic Net** regularization techniques to mitigate multicollinearity.
  - **Code**: GitHub Repository
- **Spatio-Temporal Audio Classification** *University of Georgia*
  Course Project in Machine Learning in IoT (under Prof. Fei Dou) *Jan 2024 - May 2024*
  - **Model Adaptation**: Adapted a published **ConvLSTM-based architecture** to classify underwater acoustic signals on the **VTUAD dataset**.
  - **Learning Outcome**: Gained initial hands-on exposure to deep learning pipelines, model reproduction, and working with spatio-temporal data in ML.